

Deep Learning

The objectives :

- Understand the main concepts of deep learning
- Be able to explain how deep learning is work and how applied to supervised learning.
- Understand what are the major categories of models (such as MLP, CNNs and RNNs), and when they should be applied.
- Build Deep Learning Model Using Python specially Keras Library.

Machine Learning Overview

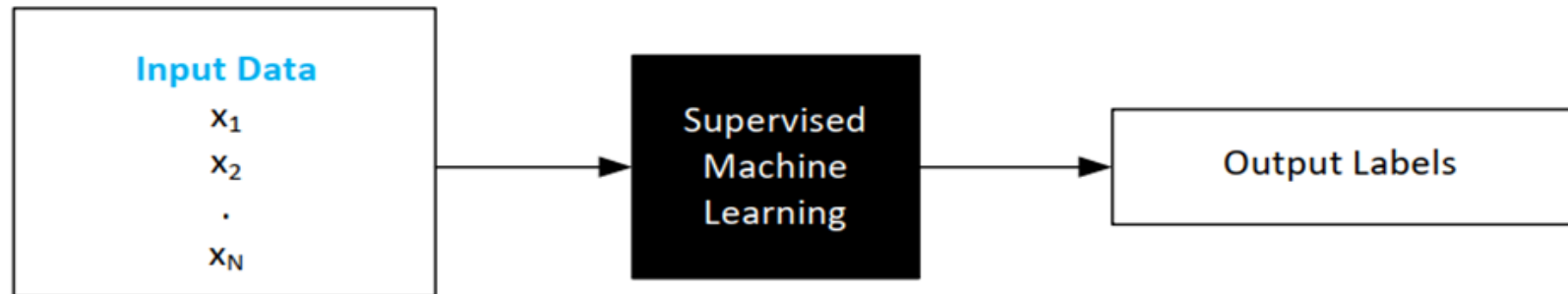
Machine learning is the study of computer algorithms that can improve automatically through experience and by the use of data. It is seen as a part of artificial intelligence.

There are two main types of Machine Learning:

1. Supervised Learning
2. Unsupervised Learning

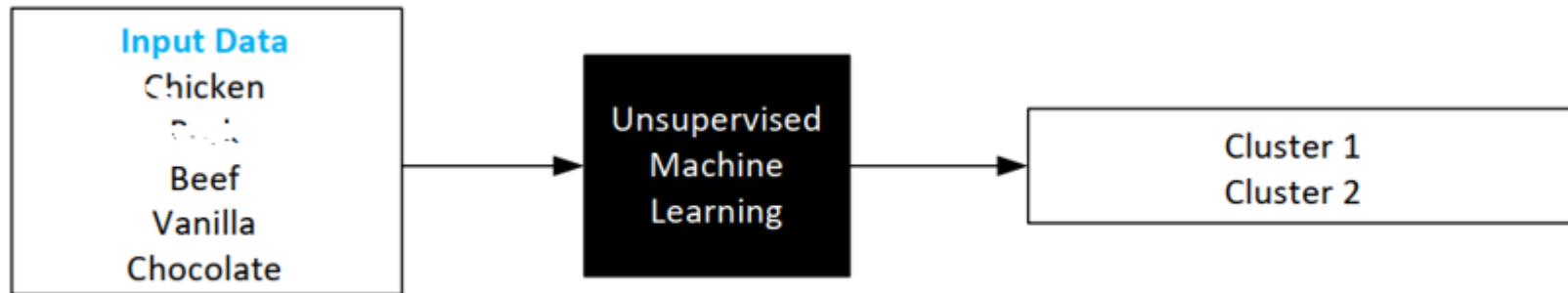
Supervised learning

- supervised learning is by far the most popular form of AI and ML used today.
- We take a set of labeled data (called a dataset) and we feed it to some ML learning algorithm that develops a model to fit this data to some outputs.
- Regression or classification ?



Unsupervised Learning

- Unsupervised learning is concerned with finding interesting clusters of input data. It does so without any help of data labeling.
- It does this by creating interesting transformations of the input data
- It is very important in data analytics when trying to understand data.



ML Supervised Learning Process

Step 1 – Obtain a labeled data set.

Step 2 – Split dataset into a training portion and validation or test portion.

Step 3 – Fit model to training dataset.

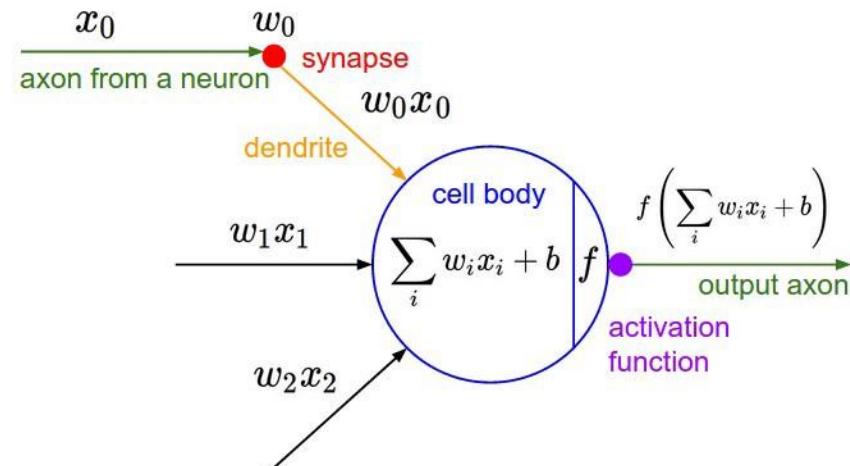
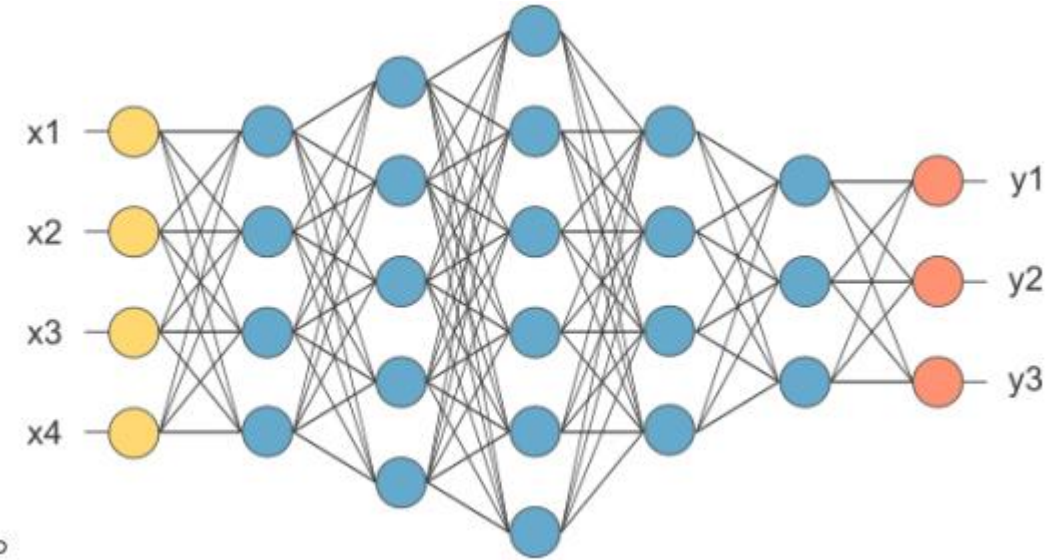
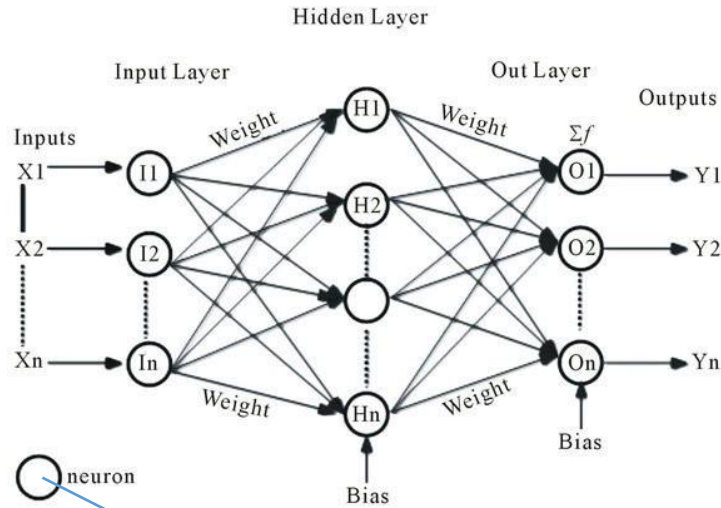
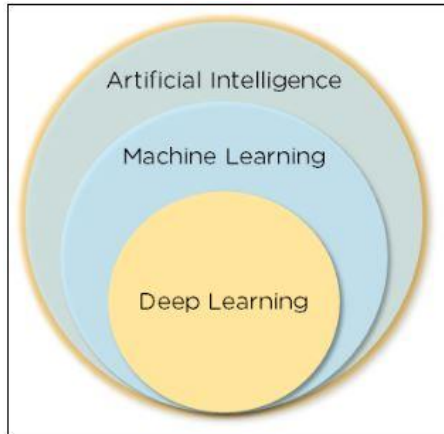
Step 4 – Evaluate models performance on the test dataset.

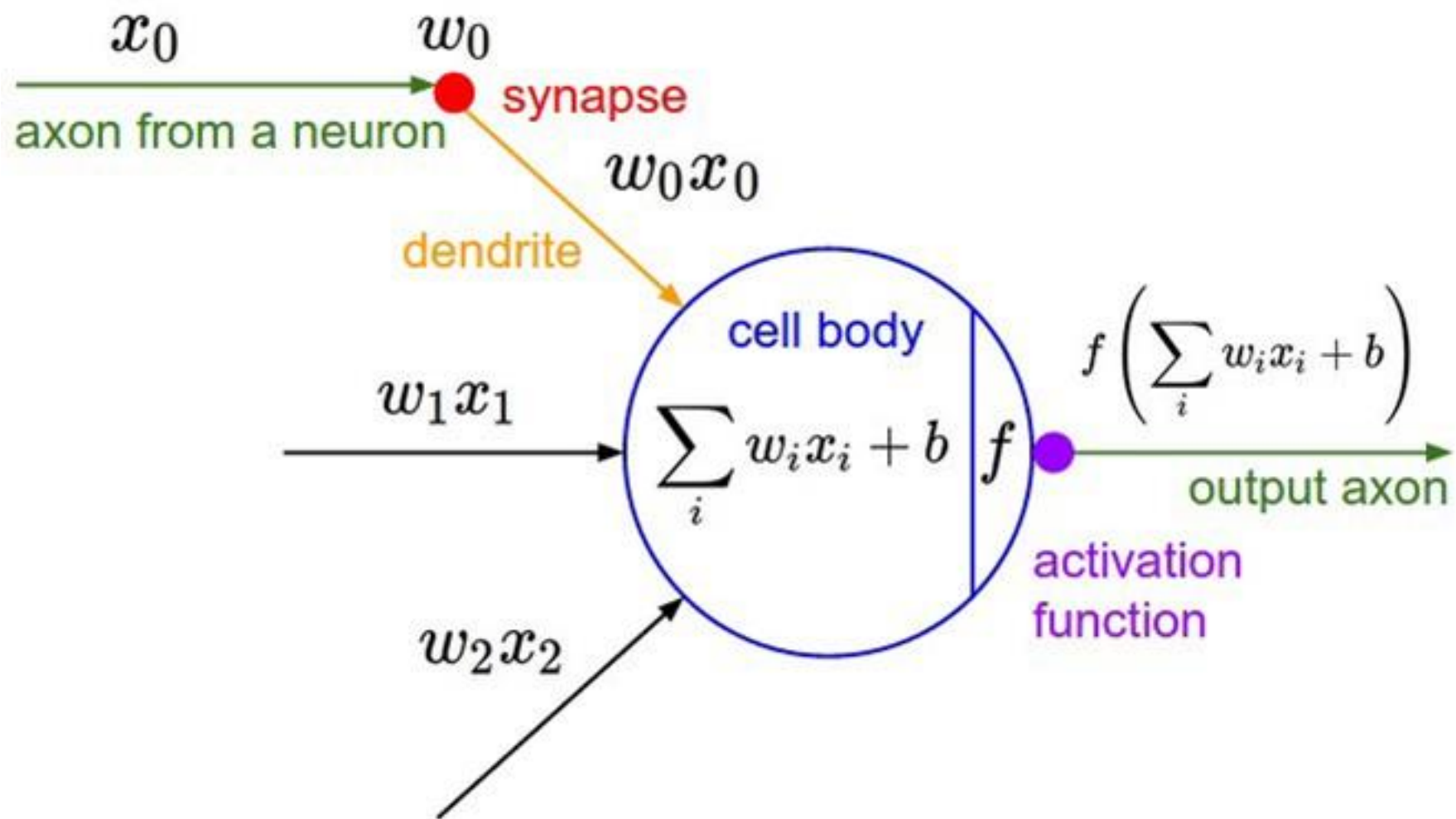
What is Deep Learning (DL) ?

Deep Learning is a subfield of machine learning concerned with algorithms inspired by the structure and function of the brain called **artificial neural networks**.

If you provide the system **tons of information**, it begins to understand it and respond in useful ways.

What the benefit for increase Depth(Layers)?



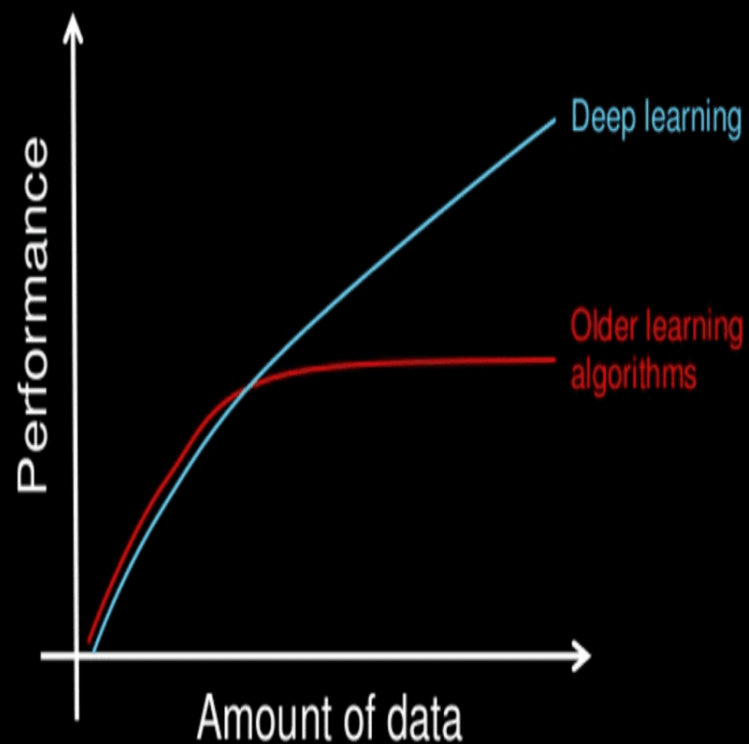


Importance of Deep Learning

- Machine learning works only with sets of structured data, while deep learning works with both **structured and unstructured data**.
- Deep learning algorithms can perform **complex** operations efficiently, while machine learning algorithms cannot. **how?**
- The **performance** of machine learning algorithms **decreases** as the number of **data increases**; so to maintain the performance of the model, we need a deep learning.

Why Now?

Why deep learning



How do data science techniques scale with amount of data?

Neural Networks date back decades, so why the resurgence?

1. Big Data

- Larger Datasets
- Easier Collection & Storage

IMAGENET



WIKIPEDIA
The Free Encyclopedia



2. Hardware

- Graphics Processing Units (GPUs)
- Massively Parallelizable

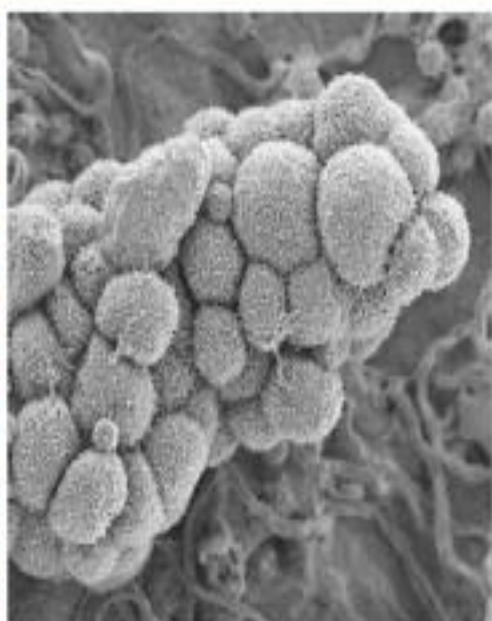


3. Software

- Improved Techniques
- New Models
- Toolboxes



DEEP LEARNING EVERYWHERE



INTERNET & CLOUD

Image Classification
Speech Recognition
Language Translation
Language Processing
Sentiment Analysis
Recommendation

MEDICINE & BIOLOGY

Cancer Cell Detection
Diabetic Grading
Drug Discovery

MEDIA & ENTERTAINMENT

Video Captioning
Video Search
Real Time Translation

SECURITY & DEFENSE

Face Detection
Video Surveillance
Satellite Imagery

AUTONOMOUS MACHINES

Pedestrian Detection
Lane Tracking
Recognize Traffic Sign

Image segmentation and recognition

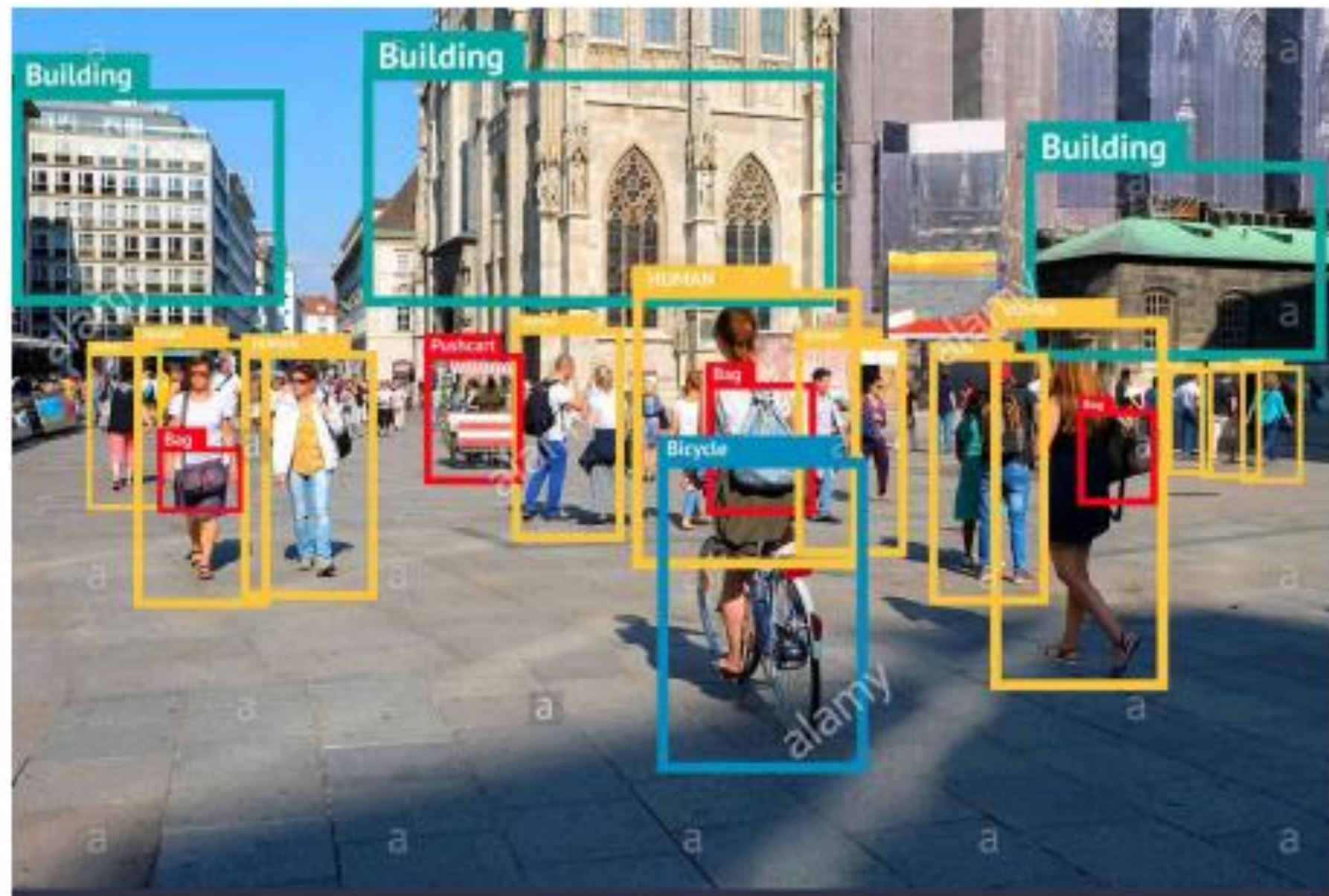
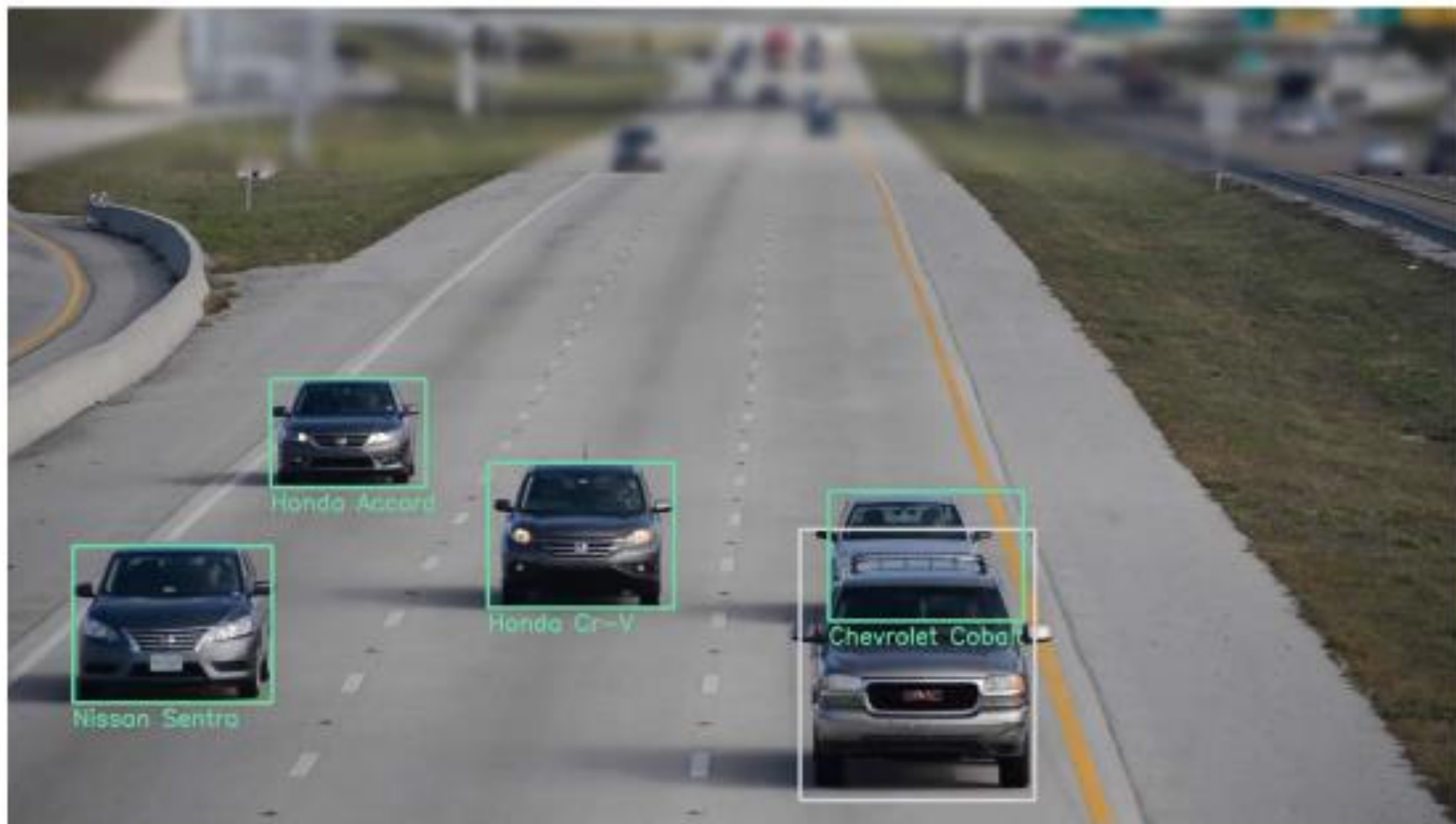


Image recognition



Breakthroughs with neural networks

ThisPersonDoesNotExist.com uses AI to generate endless fake faces

Hit refresh to lock eyes with another imaginary stranger

by James Vincent | Feb 16, 2019, 7:38am EST

f t SHARE

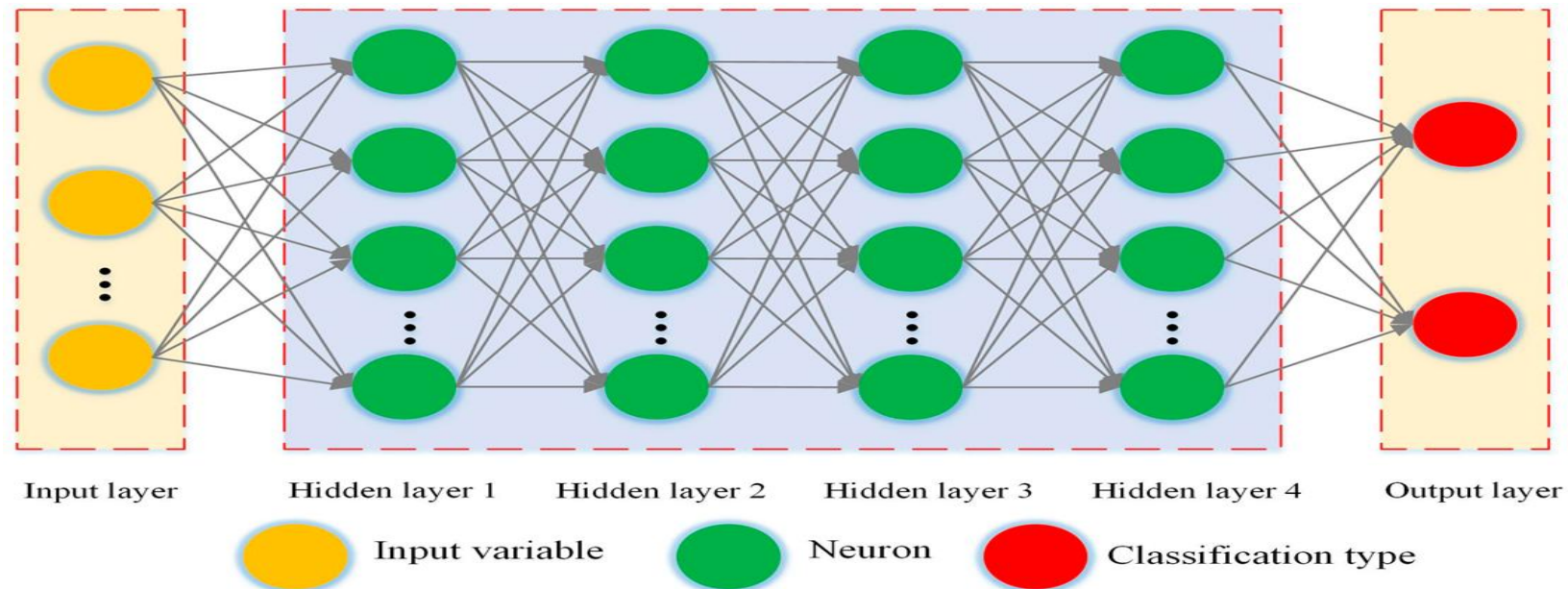


A few sample faces — all completely fake — created by ThisPersonDoesNotExist.com.

Types of Algorithms used in Deep Learning

• Multilayer Perceptrons (MLPs)

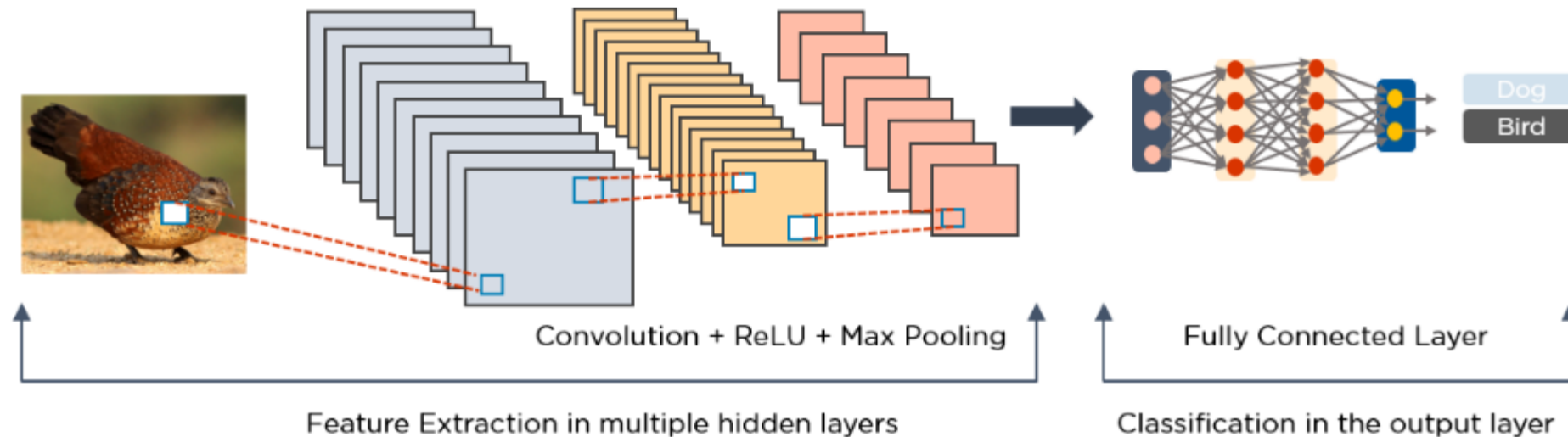
- MLPs are an excellent place to start learning about deep learning technology.
- MLPs belong to the class of feedforward neural networks with multiple layers of perceptrons that have activation functions. MLPs consist of an input layer and an output layer that are fully connected. They have the same number of input and output layers but may have multiple hidden layers and can be used to build speech-recognition, image-recognition, and machine-translation software.



- **Convolutional Neural Networks (CNNs)**

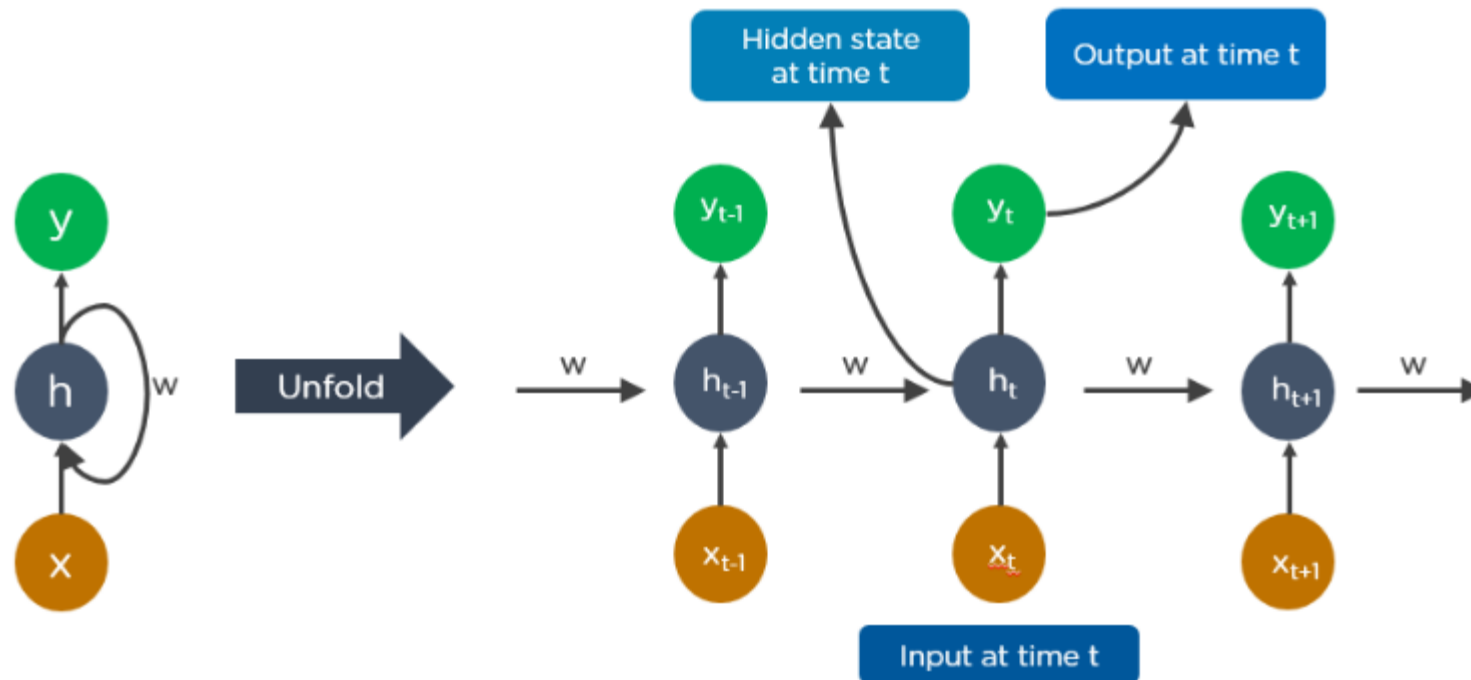
CNN's, also known as ConvNets, consist of multiple layers and are mainly used for image processing and object detection.

CNN's are widely used to identify satellite images, process medical images.



- **Recurrent Neural Networks (RNNs)**

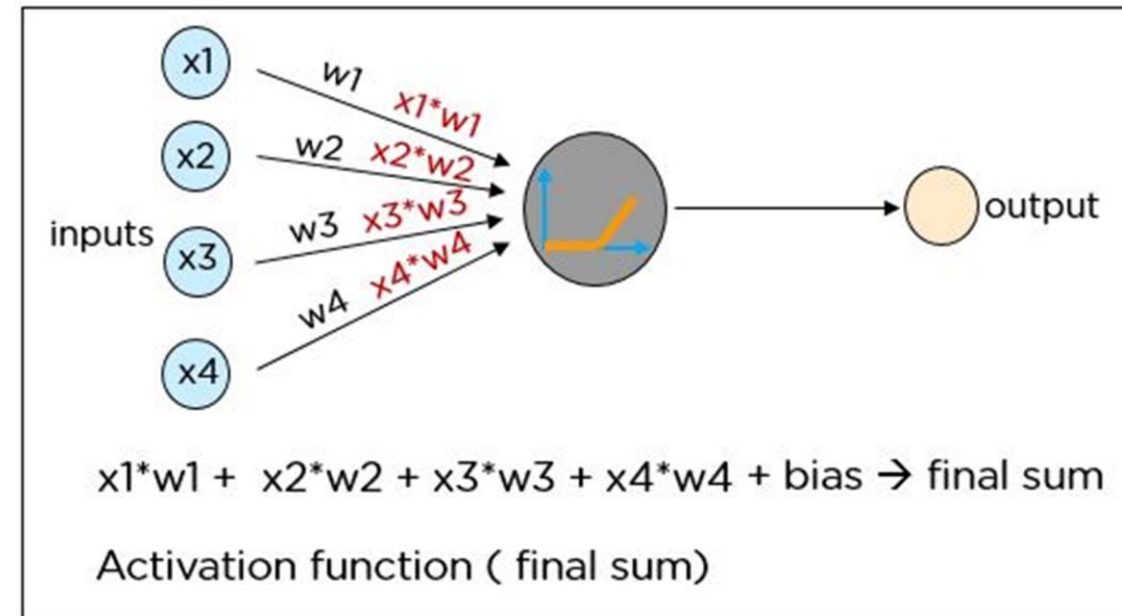
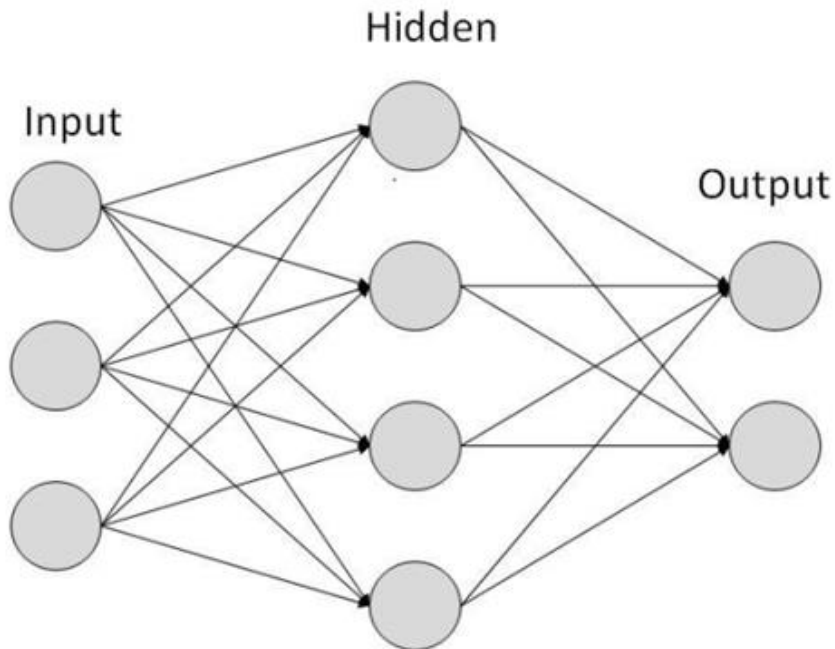
- RNNs have connections that form directed cycles, which allow the outputs from the LSTM to be fed as inputs to the current phase.
- The output from the LSTM becomes an input to the current phase and can memorize previous inputs due to its internal memory.
- RNNs are commonly used for image captioning, time-series analysis, natural-language processing, handwriting recognition, and machine translation.



Neural Networks layers

- A **neural network** is structured like the human brain and consists of artificial neurons, also known as nodes. These nodes are stacked next to each other in three layers:
- The input layer
- The hidden layer(s)
- The output layer

Data provides each node with information in the form of inputs. The node multiplies the inputs with random weights, calculates them, and adds a bias. Finally, nonlinear functions, also known as activation functions, are applied to determine which neuron to fire.



Anatomy of a neural network

- *Layers*, which are combined into a *network* (or *model*)
- The *input data* and corresponding *targets*
- The *loss function*, which defines the feedback signal used for learning
- The *optimizer*, which determines how learning proceeds

